Table 515-F: Requirements for Virgin Coarse Aggregates

ASTM Test Reference	Requirements	
C127	Maximum Water Absorption; % by Mass	2
C88	Soundness of Aggregate Maximum Loss After 5 Cycles	20
C142	Maximum % by Mass of Clay Lumps and Friable Particles	1.5
D6928	Minimum Degradation Factor	35
D5821	Two (2) Fractured Faces: Minimum % by Mass Retained on the 4.75 mm sieve	85

(b) Virgin Fine Aggregate:

(i) Shall be clean, tough, durable, moderately sharp, and free from coatings of clay, silt, or other deleterious material, and shall contain no clay balls or other aggregations of fine material.

(ii) Shall have a maximum mass loss after five cycles of not more than 23% when tested in accordance with ASTM C88.

(iii) Shall have a sand equivalent of not less than 40 when tested in accordance with ASTM D2419.

(iv) Shall have a minimum value of 45 when tested according to AASHTO T 304, Method "A", – Uncompacted Void Content of Fine Aggregate when determining Fine Aggregate Angularity.

(c) Virgin Mineral Filler and Mineral Dust:

(i) Mineral filler shall consist of all matter passing the 0.600 mm sieve and mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

(ii) Mineral filler and mineral dust shall be free from organic matter.

(iii) Mineral filler shall be non-plastic when tested in accordance with ASTM D4318.

515.05.03 Supply of Asphalt Cement and Rejuvenating Agents – The Contractor shall supply the types and grades of asphalt cement and rejuvenating agents as specified in the Special Provisions. The supply of these materials includes, but is not limited to, ordering, scheduling delivery of, supply of Temperature Viscosity Curve information, receiving, handling, storing, heating, blending, sampling, and testing of the materials and other related work.

The Contractor shall supply the Ministry Representative with copies of the supplier's weigh-bill and records of all asphalt and rejuvenating materials received on a daily basis. The amount of asphalt cement used will be calculated based on the Contractor's Quality Control testing results.

The supply of asphalt cement and rejuvenating agents shall be at the Unit Price bid for asphalt cement and rejuvenating materials.

515.05.04 Responsibility for Asphalt Admix Design – The Ministry will provide to the Contractor information related to the Asphalt Admix and rejuvenating ratios in the Special Provisions, but all admix design and formulation costs are the responsibility of the Contractor.

515.05.05 Evaluation of Asphalt Admix Materials – The Ministry Representative will require up to five (5) days upon notification from the Contractor to evaluate the Contractor's Asphalt Admix materials.

515.05.06 Samples Required for Asphalt Admix Evaluation – At the discretion of the Ministry Representative, the Ministry may request samples of the Admix materials and Quality Control results and documentation for evaluation. Shipping costs for samples delivered to the Ministry Representative are the responsibility of the Contractor.

515.05.07 Field Adjustment of the Job Mix Formula and/or Recycled Pavement – During construction, the Ministry may require field adjustments to the job mix formula and/or recycled pavement. A field adjustment to the Job Mix Formula is defined as a change in the asphalt cement content of admix, aggregate gradation, rejuvenating agent and/or proportioning of various aggregate sizes.

PAYMENT ADJUSTMENTS

515.11 Density

515.11.01 Coring – The Contractor shall be responsible for providing all core samples for quality assurance and payment adjustment purposes. The location of the cores shall be randomly selected and provided to the Contractor by the Ministry Representative. The cores shall be taken within a 300 mm radius of that location unless otherwise agreed to by the Ministry Representative. The Contractor shall provide 150 mm diameter cores for these purposes. The minimum core thickness shall be 37 mm. The Contractor shall prepare the cores prior to submission by removing all material not representative of the pavement lift to be tested. The Contractor shall deliver these cores to the Ministry Representative on-site, within 24 hours of being provided the locations for the coring.

In the case of a core sample location falling in an obvious non representative area, such as a previously patched area, at the sole discretion of the Ministry Representative, a new random location will be chosen for that sample.

All costs associated with obtaining the cores, including the filling and compaction of the core holes are considered

incidental to the contract and are the responsibility of the Contractor.

515.11.02 Percent Density – One random core sample will be obtained from each Sub-Lot and tested. The test results for three Sub-Lots will be averaged to determine the percent density for the Lot.

$$\text{Hot-In-Place \% Density} = \frac{\text{In Place Density of Sample}}{\text{Maximum Theoretical Density (ASTM D 2041)}} \times 100$$

The Maximum Theoretical Density for the Lot will be determined by combining the Sub-Lot cores. The cores shall be prepared in accordance with the procedures outlined in APPENDIX 515-C to enable the testing to determine the maximum theoretical density.

515.11.03 Payment Adjustments – The payment adjustment for percent density will be the amount shown in Table 515-G for the Sample Mean of the test results for the Lot.

515.11.04 Rejection Limit – The rejection limit for percent density is the limiting value of the Sample Mean as shown in Table 515-G.

If the test result for the density of a Sub-Lot is outside the acceptance limits, the Sub-Lot is rejected automatically regardless of the values of other acceptance parameters. To minimize the cost of rejection to the Contractor, the Contractor shall isolate the area of low density within the Sub-Lot and perform the necessary corrective measures to ensure specifications are met. The limits of the low density area must be verified and approved by the Ministry Representative before remedial work proceeds.

515.11.05 Payment for Rejected Work Made Acceptable – The payment adjustment for density will be based on testing of the reprocessed, replaced or overlaid material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work.

Table 515-G: Payment Adjustments for Density

% Maximum Theoretical Density (Lot Average)	Payment Adjustment ($ per m^2) per Lot
≥ 95.6	+ $0.50
95.0 – 95.5	+ $0.30
94.0 – 94.9	+ $0.20
93.0 – 93.9	$0.00
92.6 – 92.9	- $0.20
92.0 – 92.5	- $0.50
≤ 91.9	REJECT

515.12 Smoothness

515.12.01 Determination of Pavement Smoothness – The finished pavement surface shall be tested by the Ministry using either a Class I precision rolling profile measuring instrument or a non-contact Inertial Profiling measuring instrument, to determine the longitudinal profile and compute the International Roughness Index (IRI) in each driving lane. Profiles shall be measured and the IRI calculated in the centre of the lane for each Sub-Lot. IRI values will be recorded to a precision of 0.01 m/km for each Sub-Lot. The Lot IRI value is the average of the IRI values calculated for the Sub-Lots within the Lot.

For any Sub-Lot between 50 m and 100 m in length, the IRI value shall be considered representative of a complete Sub-Lot. For any Sub-Lot less than 50 m in length, the IRI value will be combined with the proceeding Sub-Lot IRI value.

The profile shall be measured over the entire length of the pavement exclusive of structures and shoulder areas. Acceleration, deceleration and turning lanes are considered part of the driving lanes and shall be testing in accordance with this provision. For the measuring process, the Contractor shall provide the Ministry Representative a chalk guide line in the centre of the lane immediately prior to measurement with a Class I precision rolling profile measuring instrument. Payment for providing a chalk line for a Class I precision rolling profiler will be paid upon acceptance by the Ministry Representative under the Provisional Sum Item.

515.12.02 Auxiliary Lanes – For smoothness testing, sections of the driving lanes that do not fall within the continuous through lanes, such as acceleration lanes, deceleration lanes and turning lanes, and lanes which are less than 1 km in length, shall be treated as follows. The ratio of the section length to the standard Lot length of 1 km shall be determined and the payment adjustment shall be pro-rated for the short section by that ratio as follows:

$$Ratio = \frac{Actual\ Section\ Length\ (m)}{1000\ m}$$

Hence, as an example, for an actual section length of 565 m, the applicable payment adjustment is 0.565 times the payment adjustment for a 1 km Lot as determined from Table 515-H.

515.12.03 Acceptance Limits – The acceptance limit for smoothness is the limiting value as shown in Table 515-H, beyond which corrective work is required. Payment adjustments are shown in Table 515-H. The International Roughness Index (IRI) value, calculated for each Sub-Lot, will be used to determine if the Lot will be accepted, and if so whether it will be subject to any payment adjustments.

515.12.04 Payment Adjustments for Full or Increased Payment – Acceptance of any Lot at full or increased payment will occur if:

(a) it contains no obvious defects as per SS 515.22 and

(b) in the case of top lift pavement only, the pavement smoothness, as reported by the International Roughness Index (IRI), for each Sub-Lot shall have an IRI value ≤ 2.0

515.12.05 Rejection Limit – If a Sub-Lot has an IRI >2.0, the Sub-Lot is rejected.

515.12.06 Remedial Work – If the test results on a Sub-Lot of pavement indicate a payment reduction or rejection because of smoothness, the Contractor may propose remedial work to improve the smoothness. Such proposals are subject to the approval of the Ministry Representative, but such approval does not imply that the proposed remedy will be successful and does not reduce the Contractor's responsibility for meeting the acceptance requirements. Reprocessing may be acceptable, but cold milling and repaving may be required. Only one attempt to remediate the reject area (remediation method chosen by the Contractor) may be made to improve smoothness, and this must be completed within ten (10) days from the time the Contractor receives notification from the Ministry Representative of the original smoothness test results for the Sub-Lot. Should a product still be in reject the Ministry Representative will determine the final remediation method at the Contractor's expense.

No payment will be made for any material, equipment or manpower used to improve, or attempt to improve, smoothness.

515.12.07 Smoothness Deficiency Report – Smoothness deficiencies (bumps and dips) less than 8 mm over 3 m will not have a fix or remedial work requirement. Individual smoothness deficiencies between 8 mm and 12 mm over 3 m will result in a $400.00 penalty per deficiency and smoothness deficiencies over 12 mm over 3 m will require remedial work (see SS 515.22).

Table 515-H: Lot Assessment and Payment Adjustments for Smoothness

Lot IRI (m/km)	Payment Adjustment
≤1.00	+$2,500
>1.00 and ≤1.10	+$1,200
>1.10 and ≤1.20	+$700
>1.20 and ≤1.30	+$400
>1.30 and ≤1.40	0
>1.40 and ≤1.60	-$500
>1.60 and ≤1.80	-$900
>1.80 and ≤1.90	-$2,000
>1.90 and ≤2.00	-$2,500
>2.00	REJECT

ACCEPTANCE AND REJECTION

515.21 Acceptance at Reduced or Adjusted Payment – Acceptance of any Lot at reduced payment will occur if it contains no obvious defects as per SS 515.22, and if:

(a) The test results for acceptance parameters are such that the Lot or Sub-Lot meets the requirements for acceptance at a reduced payment;

(b) The Lot and Sub-Lot is approved in respect of all other requirements; and

(c) The Contractor has not notified the Ministry Representative in writing that it will exercise its option to either repair or remove and replace the work, at its own cost, with work meeting the requirements for acceptance at full or increased payment.

515.22 Rejection for Workmanship Defects – The finished surface of any lift shall have a uniform "closed" texture and be free of visible signs of poor workmanship. Any obvious defects as determined by the Ministry Representative such as, but not limited to the following, will cause for automatic rejection of asphalt pavement regardless of the values of any other acceptance parameter:

(a) Individual bumps and dips that exceed 12 mm over 3 m;

(b) Areas of excess or insufficient asphalt;

(c) Improper matching of longitudinal and transverse joints;

(d) Roller marks;

(e) Tire marks;

(f) Segregation; or

(g) Cracking or tearing

When asphalt pavement is rejected by reason of obvious defects, the minimum area of rejection will be the actual length of the defect for the full width of the driving lane in which the defect exists.

Rejected work shall be promptly repaired, remedied, or removed and replaced in a manner acceptable to the Ministry Representative. The Contractor shall be responsible for all costs including materials.

No payment will be made for work in any Lot or Sub-Lot, which has been rejected, until the defects have been remedied.

515.23 Appeal Testing

515.23.01 Density – The Contractor may appeal the results of acceptance testing for density for any Lot only once. Appeals will only be considered if quality control results can be presented to support the appeal.

Quality control test results for density which are provided to the Ministry Representative subsequent to the Contractors

receipt of the quality assurance test results for that Lot will not be considered (when evaluating evidence) for an appeal. The appeal shall be for all tests within the Lots, and there will be no appeal allowed for single test within the Lot.

The following procedures will apply for the appeal:

(a) The Ministry Representative will arrange for an independent testing laboratory to perform the appeal testing. The personnel employed or testing laboratory retained by the Contractor for quality control testing on the project will not be used for appeal testing.

(b) The original quality assurance results will not be considered in the appeal. The Contractor will sample the pavement at locations randomly re-selected by the Ministry within five (5) days following receipt of the appeal such that a total of five (5) new core specimens are extracted throughout the Lot. The Ministry Representative will observe the sampling process. The procedures used to determine the test results shall be consistent with the original acceptance testing methods.

515.23.02 Smoothness – The Contractor may appeal smoothness acceptance test results of any rejected or penalized Lot, once. The appeal shall be in writing and submitted within the next five (5) days following receipt of the test results.

Any attempt to improve smoothness on the appealed Sub-Lot after the Ministry Representative has tested the Lot for acceptance shall void the appeal and the original test results will apply.

The Ministry Representative will perform, and the Contractor will be given the opportunity to witness, the appeal testing and the new results will be binding on the Contractor and the Ministry.

515.23.03 Application of Appeal Testing Results – For density appeals, the original test will be discarded. A new Sample Mean for the five (5) new test results will be determined and used for acceptance and payment adjustment.

The new values, thus determined, in all cases, will be binding on the Contractor and the Ministry.

515.23.04 Payment for Appeal Testing – If the new results indicate a change in the payment adjustment in the Contractor's favour, then sampling and testing costs incurred during appeal procedures for that Lot will be borne by the Ministry.

If the new results verify that any payment reduction or rejection remains valid for that Lot, then the costs of sampling and testing (plus 10% mark-up) incurred during the appeal procedure will be charged to the Contractor.

515.23.05 Time Limits for Appeals of Test Results – All appeals shall be in writing and submitted within the next three (3) site occupancy days of receipt of the test results.

515.24 Cold Milling and Re-Paving as a Corrective Measure – If cold milling and re-paving is used as a corrective measure on a defective Lot or Sub-Lot, the thickness will be subject to the approval of the Ministry Representative, but shall not be less than 40 mm. In all other respects, the re-paving will be subject to the same specifications as the pavement being replaced.

Whether the cold milling and re-paving is applied as a corrective measure, acceptability and payment will be determined as follows:

(a) Acceptability, and eligibility for either positive or negative payment adjustment, will be determined entirely on the results of testing and observations conducted on the re-paving, regardless of test results that have been obtained on the Hot-In-Place recycled pavement.

(b) The payment quantity, for application of the Unit Prices for asphalt pavement, and the quantity, to which any payment adjustment is to be applied, will be derived from the square metres of recycled pavement affected by the re-paving.

PAYMENT

515.31 General – Payment for Hot-In-Place Recycle of asphalt pavement will be at the Contract Unit Price per Square Metre. The Contract Unit Price shall be for all things furnished and done including sweeping, processing, compacting, patching and any hand work as necessary to complete the work. Payment at Unit Prices for supply of virgin aggregates, the supply of asphalt cement, the supply of rejuvenating agents, the supply of virgin asphalt mix and the Hot-In-Place recycling construction of asphalt pavement, shall be full compensation for completing the supply and installation of Hot-In-Place asphalt pavement on prepared surfaces in accordance with the contract requirements. Applicable payment adjustments (additions or subtractions as applicable) shall be applied in accordance with the PAYMENT ADJUSTMENT Section.

The first 4,000 m^2 of Hot-In-Place asphalt pavement recycling production will not be subject to bonus/penalty payment adjustments but will be required to meet minimum specification requirements for quality and workmanship.

515.31.01 Rejuvenating Agent and Asphalt Cement

(a) **Rejuvenating Agent** – Payment to Supply Rejuvenating Agent shall be at the Unit Price per litre for the quantity of material actually metered into the recycling process. The Contract Unit Price shall be to supply, store, heat, apply and proportioning of the agent with the asphalt pavement mixture reclaimed from the roadway.

(b) Asphalt Cement – Payment for Supply of Asphalt Cement for admix shall be at the Unit Price per tonne for the payment quantity determined from

$$\text{Payment Quantity of Asphalt Cement} = \frac{\text{Av. Actual Asphalt Content X Payment Quantity of Asphalt Admix}}{\text{Average Actual Asphalt Content} + 100}$$

Where:

Payment Quantity of Asphalt Admix = quantity determined in accordance with SS 515.31.03,

Actual Asphalt Content = defined in SS 515.03(d),

The actual asphalt content shall be determined on a Lot by Lot basis by averaging the value obtained from the Quality Control tests from each Lot. This average will be used to determine the tonnage of Asphalt Cement used for payment purposes for the Asphalt Cement for the Lot.

The quantity of asphalt cement so determined shall be used for payment purposes regardless of any discrepancy that may be noted between it and any quantity calculated using delivery quantities, tank dips, or other data. No payment will be made for asphalt cement used in reject mix.

Such payment shall be full compensation for supplying the material to the project, handling, storing, heating, sampling, and testing of the material, and other related work.

515.31.02 Paving Aggregate into Stockpile (EPS) – Payment for Paving Aggregate into stockpile (EPS) shall be at the Unit Price per tonne for the quantity of paving aggregate placed into stockpile in accordance with the following:

(a) The paving aggregate quantity will be identical to 100% of the asphalt admixture quantity.

(b) As the aggregate is crushed into stockpile, progress payments will be made against the bid item, up to the quantity shown in the "Approximate Quantity" column of the Schedule 7.

(c) At contract completion, the final payment quantity for asphalt admix aggregate will be in accordance with the following calculation:

$$\text{Tonnes of Asphalt Admix Aggregate} = \frac{\text{tonnes of mix X 100}}{100 + \text{actual calculated asphalt content as per SS 515.03(d)}}$$

Such payment shall be full compensation for all work include, but not limited to, the production, supply and stockpiling of all paving aggregates, and shall include all costs of quality control.

515.31.03 Asphalt Admix (EPS) – Payment for Asphalt Admix (EPS) constructed in place shall be at the Unit Price per tonne for the quantity of admix placed in accordance with the Contract requirements.

Subject to the exception noted below, only acceptable asphalt admix will be included in the payment quantity.

Where cold milling and re-paving is used as a corrective measure, in accordance with SS 515.24;

(a) The overlay quantity will not be included in the admix payment quantity; but

(b) The admix used in the quantity of recycled pavement removed by the cold milling and re-paving will be included in the admix payment quantity, whether or not it was acceptable.

Such payment shall be compensation in full for all work including but not limited to loading the aggregate into the feeders, drying the aggregate, metering and adding the asphalt cement, mixing, loading, weighing, hauling, dumping, spreading, compacting and finishing the asphalt pavement, and shall also include all costs of quality control.

515.31.04 Hot-In-Place Recycled Asphalt Pavement (EPS) – Payment for Hot-In-Place Recycled Asphalt Pavement (EPS) constructed in place shall be at the Unit Price per square meter for the quantity of recycled pavement in accordance with the Contract requirements.

Subject to the exception noted below, only acceptable recycled asphalt pavement will be included in the payment quantity.

Where cold milling and re-paving is used as a corrective measure, in accordance with SS 515.24;

(a) The overlay quantity will not be included in the admix payment quantity; but

(b) The admix used in the quantity of recycled pavement removed by the cold milling and re-paving will be included in the admix payment quantity, whether or not it was acceptable.

Such payment shall be compensation in full for all work including, but not limited to recycling, metering, adding the rejuvenating agent, adding, spreading, compacting and finishing the asphalt pavement and shall also include all costs of quality control.

515.31.05 Payment for Acceptable Work – The following end product properties of recycled asphalt pavement will be measured for acceptance:

(a) Density

(b) Smoothness

515.31.06 Payment for Rejected Work Made Acceptable – When defects have been remedied in Lots or Sub-Lots which had been rejected, payment for the original quantity of material in those Lots or Sub-Lots will be made subject to payment adjustments and penalty assessments and subject to SS 515.24.

No payment will be made for any material used to replace, repair or overlay rejected work and all corrective work shall be performed entirely at the Contractor's expense.

515.31.07 Payment Adjustment – Payment adjustments resulting from the application of this Section will be affected on each progress payment as follows:

(a) For each Lot paid for by the square metre (m^2), the applicable payment adjustment derived from Table 515-G (Density), in dollars per square metre will be expressed as positive in the case of increases and negative in the case of decreases. The algebraic sum of these unit adjustments will then be applied to the payment quantity for the Lot. The resulting amount, in dollars, will be the net payment adjustment, positive or negative, for that Lot.

The algebraic sum of the net payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for density for the current process payment.

(b) For each Lot, the applicable payment adjustment derived from Table 515-H (Smoothness), in dollars per Lot, will be expressed as positive in the case of increases and negative in the case of decreases, and will be the payment adjustment, positive or negative, for that Lot.

The algebraic sum of the payment adjustment for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for smoothness for the current progress payments.

(c) The algebraic sum of the total payment of the total payment adjustments for density and smoothness, derived in accordance with a) and b) above, shall be the total payment adjustment, positive or negative, in dollars, for all attributes for the current progress payment. This amount shall be added, if positive, or subtracted, if negative, by a single entry in computing the current progress payment.

(d) The process set out in SS 515.31.07(a) to SS 515.31.07(c) above will be used in computing each progress payment to which it is applicable.

515.32 Ministry Purchase of Surplus Aggregate in Stockpile

515.32.01 Surplus in Private Pits – At the discretion of the Ministry Representative, the Ministry may or may not purchase surplus aggregate in stockpile in a private pit.

Generally, the Ministry will provide payment for the processing costs of surplus aggregate stockpile in Ministry pits only, and only to a limited quantity.

However, should the Contractor produce surplus aggregate in a private pit, and the Ministry intends to purchase these surplus aggregates, the Contractor shall be required to provide a written agreement with the owner of the property. This document shall indicate that the Ministry will have free access to and use of the surplus aggregate in stockpile for a period of 12 months after the completion of the contract work.

If the Contractor undertakes private work from within the private pit, measurements for surplus aggregate in stockpile will not be taken until the completion of the private work, ensuring that the Ministry does not pay for aggregate used on private works.

All surplus aggregate shall be properly stockpiled.

515.32.02 Ministry Purchase of Surplus Aggregate – Should the Ministry proceed with the purchase of surplus aggregate, upon completion of the contract, the Ministry will purchase surplus paving mix aggregate as indicated herein.

If the quantity of Asphalt pavement actually incorporated into the works is less than the estimated quantity, as stated in the Schedule 7, the Ministry will purchase aggregate up to 100% of the quantity required by the contract, to be paid under the appropriate Unit Price Item for Aggregate in Stockpile.

The surplus aggregate must, when singly or combined, meet the gradation requirements set out in the Job Mix Formula.

The portion of material remaining in a stockpile eligible for compensation as surplus aggregate shall be the lesser of:

(a) The quantity of mix as stated in the Schedule 7 minus the quantity of aggregate actually incorporated in the works; and

(b) The volume of the remaining stockpile as determined under SS 502.41.03.

No payment shall be made for any surplus shoulder and paving aggregate remaining at contract completion in excess of the contract requirement. Material remaining in a Ministry pit is the property of the Ministry.

515.32.03 Measurement of Surplus Aggregate – The Ministry will determine the volume of aggregate by surveying the stockpile(s) using string-line techniques, and determine volume using prismoidal volumes between surfaces. Alternatively, the Ministry Representative may elect to survey using cross section techniques and/or determine volumes using end-area volumes.

The base of the pile will be determined from a pre-stockpile survey or, where such a survey is not available, from a surface determined by the Ministry Representative as being a reasonable interpolation from the intersection of the pile sides with the adjacent ground level. Where there are two or more stockpiles of aggregate meeting the same gradation classification, the compensation will be based on the cumulative quantity of those stockpiles.

Where necessary, stockpile volume shall be converted to mass using the conversion factor of 2.0 tonnes per cubic metre.

No payment shall be made for any surplus aggregate remaining at contract completion in excess of the contract requirement. Material remaining in the pit is the property of the Ministry.

Surplus shouldering, surfacing, crushed base and subbase aggregates will be paid in accordance with SS 202.

APPENDIX 515-A

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

515-A.01 GENERAL – The Contractor's responsibility for preparing submitting and adhering to a Quality Control Plan are specified in SS 515.04. This Appendix provides requirements and guidelines for the Contractor's Quality Control Plan, in addition to those set out in SS 515.04.03.

515-A.02 QUALITY CONTROL PLAN – As a requirement, the Contractor shall prepare and submit a Quality Control Plan for evaluation. The Plan may be operated wholly, or in part by a qualified Subcontractor or an independent organization/agency. However, the Quality Control Plan, including compliance with the Plan and its modifications, is the responsibility of the Contractor.

515-A.02.01 Plan Elements – The Plan shall address all elements that affect the quality of the Hot-In-Place recycled asphalt pavement including, but not limited to, the following:

(a) Supply of Asphalt Materials;

(b) Supply of rejuvenating agents;

(c) Identification of source and proof of quality of virgin aggregates to be supplied;

(d) Virgin aggregate production and its gradation control;

(e) Quality of virgin aggregate components;

(f) Stockpile management;

(g) Asphalt plant calibration and proportioning of aggregate components;

(h) Mixing including asphalt cement content control;

(i) Process temperature controls;

(j) Admix Material and Rejuvenating Agent application rates and processed dimensions, placing and finishing;

(k) Joints;

(l) Density;

(m) Smoothness; and

(n) Segregation

515-A.02.02 Additional Elements – The Plan shall also include the following:

(a) The name of the Quality Control testing agency and its proven capability to provide the specific services required for the project.

(b) The list of dedicated technical staff, if available, (including names, qualifications and relevant experience) and their proposed roles.

(c) The list of testing equipment available for project work.

515-A.02.03 Designation of Personnel – The Quality Control Plan shall include the designation of specific personnel responsible for specific quality control duties.

(a) Quality Control Manager – There shall be a designated Quality Control Manager, as per SS 515.04.03. The Quality Control Manager shall be qualified as per SS 515.04.03. The Quality Control Manager shall be responsible for the preparation and sign off of the Quality Control Plan, responsible for all Quality Control testing, and inspections, responsible for the sign off of all quality control testing and inspection records and responsible for all quality control submissions to the Ministry. The individual shall be identified and named in the Quality Control Plan submission. If the Contractor requests a change in the Quality Control Manager, then a formal request to the Ministry Representative shall be made and the Ministry will review the request for approval.

(b) Process Control Technician – There may be a designated Process Control Technician (PCT) who would be responsible to ensure the laboratory test results and other quality control practices are used to control the quality of aggregates and other mix components and to adjust and control mix proportions to meet the mix design(s). The PCT is responsible for ensuring that testing equipment, utilized for proportioning and mixing are calibrated and in good working order. The Plan may describe how the PCT's duties, including sampling methods and responsibilities are to be accomplished and documented. The Plan should also describe the criteria to be used by the PCT to correct or reject unsatisfactory materials.

(c) Pavement Quality Control Technician – There may also be a Pavement Quality Control Technician (PQT) who would be responsible to ensure that delivered materials meet the requirements of the specifications. In addition, the PQT shall be responsible for periodically inspecting all equipment used in recycling, placing, finishing, and compacting to assure its proper operating condition and to assure that recycling, placing, application rate, finishing, joint construction, and compaction is in conformance with this specification and the contract requirements.

515-A.03 CONTRACTOR'S RECORD OF QUALITY CONTROL TESTING – Test results shall be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control, both during aggregate production and also during the asphalt mixing operation, should be kept on the form H0295.

515-A.04 MATERIALS APPLICATION RATE – The Contractor shall control the Material Application Rate by monitoring the amount of asphalt admix delivered to the road against the area covered by checking the application rate every five loads.

The Contractor shall monitor the addition of rejuvenating agents by calculating the application rate, through comparing the metered amount with the total area processes every one-half hour.

The Contractor is to advise the Ministry Representative on an ongoing basis of the application rate.

515-A.05 DENSITY – The Contractor shall take core samples to determine actual pavement density. At the start of paving, the Contractor may take a minimum of two pavement cores from each Sub-Lot. The Contractor may employ a nuclear moisture-density gauge to ensure intermediate density control.

515-A.06 OTHER QUALITY CONTROL PROCEDURES – The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in the Quality Control Plan. Procedures may also be introduced after the start of work as necessary as amendments to the Quality Control Plan.

515-A.07 QUALITY CONTROL TESTING FREQUENCY – Test frequency guidelines for Quality Control are described in Table 515-I.

Table 515-I: Test Frequency Guidelines

Activity	Test	Minimum Frequency
Test During Crushing (Virgin Aggregate)	ASTM C136: Dry Sieve Analysis of Aggregate	• Split Stockpiles – One (1) for each stockpile for every Two (2) hours of production. • One main stockpile – Every 300 tonnes. • Blend Sand – One (1) for every 100 tonnes during stockpiling. • Natural filler – One (1) for every 50 tonnes during stockpiling.
	ASTM D5821: Determining the Percentage of Fractured Particles	• Every second coarse aggregate sieve test.
	ASTM C117: Sieve Analysis of Aggregates by Washing (Field Lab)	• One (1) per day on reduced sample obtained from combined samples from the crusher.
Tests During Asphalt Plant Mixing (Admix)	ASTM C136: Dry Sieve Analysis of Aggregate	• One of combined aggregate every 100 tonnes.
	ASTM C566 & ASTM D2216: Moisture Content	• Aggregate – Two (2) tests per Lot. • Asphalt mix – One (1) on first Sub-Lot and every second day.
	ASTM C117: Sieve Analysis of Aggregates by Washing (Field Lab)	• One (1) per shift on reduced sample obtained from combined samples from the plant cold feed.
	ASTM D6307: Asphalt Extraction Test – Ignition Method	• Two (2) per Lot.
	ASTM D5: Penetration of Bituminous Materials	• One (1) per Manufacturer's Batch.
	ASTM D2171: Viscosity	• Contractor's Option.
Rejuvenating Agents Tests	Rejuvenating Agent	• Contractor's Option.
Test During Asphalt Paving for Density Testing (Hot-In-Place Recycled Pavement)	ASTM D1559 or AASHTO T245: Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	• Contractor's Option.
	ASTM D2726: Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures (Briquette or Cores)	• One (1) per Sub-Lot.
	ASTM D2041: Theoretical Maximum Specific Gravity of Bituminous Paving Mixtures	• One (1) – see Appendix 515-C for Combining Cores.
	Core Samples	• One (1) core for each Sub-Lot. All cores to be 150 mm diameter.

APPENDIX 515-B

REQUIREMENT FOR PRODUCTION OF ADMIX AGGREGATES

515-B.01 GRADATION MAXIMUM PERMISSIBLE VARIATION LIMIT – The variation limits specified in Table 515-J, when applied to the Ministry specified design Admix gradation, shall not results in a gradation that is outside of the limits as specified by the Ministry.

Table 515-J: Maximum Permissible Variation Limits

Sieve Size (mm)	Maximum Variation
19.0 or 16.0	0%
12.5	± 3.5%
9.5	± 3.5%
4.75	± 3.0%
2.36	± 3.0%
1.18	± 2.0%
0.600	± 2.0%
0.300	± 1.5%
0.150	± 1.0%
0.075	± 0.75%

APPENDIX 515-C

METHOD TO COMBINE ROAD CORES INTO A SINGLE COMBINED SAMPLE FOR THE DETERMINATION OF MAXIMUM THEORETICAL DENSITY

515-C.01 Introduction – After the individual Density for each core has been determined, the Sub-Lot road cores shall be combined into one single sample for the determination of maximum theoretical density.

515-C.02 Sample Preparation

(a) The upper recycled portion of the cores shall be separated from other pavement layers by sawing or other effective methods.

(b) The minimum combined sample size required shall be 2,000 grams and 2,500 grams for 19.0 mm and 25.0 mm maximum aggregate size, respectively.

(c) Select a core trimmer to ensure that the minimum sample size is obtained depending on the thickness of the samples; 100 mm, 114 mm, or 127 mm core trimmers may be used.

(d) Place the core samples in a pan and heat to 130oC ± 5oC for about 20 minutes to allow the specimens to be easily trimmed, do not overheat. The cores can be stacked on top of each other and trimmed together or individually.

(e) Select a core trimmer to attain the minimum combined sample size. Heat the base of the trimmer by placing on a burner for a few minutes.

(f) Centre the heated trimmer on the heated cores. Allow equal clearance on all sides of the core.

(g) Hold the trimmer vertically by its handle, then press it down through the core until it reaches the bottom of the pan. Apply a slight twist if the heated core(s) are difficult to penetrate.

(h) Remove the outside cut rock mix by means of a spatula spoon and discard.

(i) Thoroughly mix the resultant combined sample prior to testing to ensure uniform aggregate coating.

(j) The complete combined sample must be tested. If the resultant sample size exceeds the available capacity of the container used for vacuum saturation, it may be tested a portion at a time.

SECTION 521

ASPHALT STABILIZED BASE COURSE MIXED IN PLACE

DESCRIPTION

521.01 Scope – This Section describes the materials, plant, equipment and work practice required for the in-place stabilization of granular base course material mixed with selected bituminous binder.

EQUIPMENT

521.21 General Condition of Plant and Equipment

(a) **Equipment List –** The Contractor shall, prior to commencing construction, submit a list of equipment available and ready for use on the Contract, and shall further certify that the equipment listed meets the requirement of this Section in every respect.

(b) **Equipment Inspection –** Before any work is permitted to commence, the Contractor's equipment and plant will be carefully inspected; should any of it fail to meet the required standards, work will not be permitted to commence until such deficiency is corrected.

(c) **Equipment Condition –** All equipment and plant shall be in good mechanical condition and adjustment and be capable of efficiently performing the work required of it.

(d) **Equipment Release –** Once the Contractor's equipment list has been accepted, provision and maintenance of all equipment listed shall become an essential part of the contract and no equipment or plant may be released from the Contract work for use elsewhere until completion, except by the express written permission of the Ministry Representative.

(e) **Access and Adjustment –** The Ministry Representative shall have access at all times to any plant, equipment or machinery to be used on this contract in order to check calibration, control or operating adjustments. Where adjustment appears necessary, a competent technician or instrument repair laboratory shall do it.

(f) **Equipment Supply –** Plant, machinery and equipment specified hereafter shall represent a minimum requirement for the type of machine described and does not represent a comprehensive list of equipment to be supplied by the Contractor, who shall supply whatever plant and equipment may be necessary to the proper and efficient completion of the work.

(g) **Skilled Operators –** All plant and equipment shall be under the control of experienced and skilled operators.

521.21.02 Pressure Distributor – The pressure distributor used for applying bituminous materials shall consist of a fully insulated tank with a minimum capacity of 5000 L permanently and rigidly mounted on a truck or trailer provided with pneumatic tires and propelled by a power unit capable of maintaining a constant and uniform speed.

521.21.03 Minimum Equipment – In order to be acceptable, the distributor shall be provided with the following minimum equipment:

(a) Quick opening gate in the dome.

(b) A measuring dipstick calibrated in 100 L increments or in 20 mm gradations and provided with a calibration chart indicating the number of litres for each 20 mm in depth of contained bituminous material for accurate checking of the distributor contents. In addition, an external and readily visible depth gauge or levelometer indicating the asphalt content of the tank to the nearest 100 L and having an accuracy of ± 50 L**.**

(c) An efficient and positive means of heating the bituminous contents uniformly to any temperature up to 175°C and maintaining the contents constantly at any selected temperature without any local overheating.

(d) An approved industrial type thermometer so situated that its face shall always be readily visible to the spray bar operator. The sensing element of the thermometer shall be placed either directly in contact with the hot bituminous material or enclosed in a thin walled well of approved manufacture, so placed as to accurately measure the temperature of the material in the tank. The thermometer shall have a minimum range of 10°C to 175°C subdivided into 1°C graduations and be accurate to ± 0.5°C.

(e) Rear mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment. The spray bars shall be capable of adjustment to provide a minimum spraying width of 1.8 m and a maximum spraying width of 7.5 m.

The feed to the spray bars must be so arranged that there shall be a constant circulation of hot bituminous material to the bars such as will ensure a uniform viscosity and constant pressure of the bituminous material at each nozzle, both before and during spraying operations. Feed manifolds to spray bars shall be provided with strainers to prevent clogging of bars and nozzles.

(f) Rotary adjustable, spray bar nozzles which shall be so designed and set as to ensure a uniformly fan shaped spray without atomization, overlapping on the surface

to be sprayed, such that there shall be produced a uniformly sprayed surface to clearly defined edges of the sprayed lane without any reduction of the rate of application adjacent to the edges of the lane. Spray nozzles shall be provided with valves capable of instant full opening or positive cut-off.

(g) A connection to a circulating pressure manifold to which a hose may be attached for a single nozzle, hand operated outlet.

(h) A pressure gauge accurate to 15 kPa and calibrated pressure pump, equipped with a tachometer registering litres per minute and driven by an independent mechanically governed power unit, so designed as to be capable of applying through the spray nozzles fitted, and accurately measuring quantities of bituminous material at a minimum rate of 900 L per minute.

(i) Both pressure gauge and tachometer shall be mounted to be readily visible to the operator.

(j) A fifth wheel tachometer, registering metres per minute and so calibrated and mounted as to be readily visible to the driver of the distributor, to enable the operator to maintain the constant speed required for application of bituminous material to the road at the specified rate.

521.22 Blade Graders – Self-propelled blade graders mounted on pneumatic tires shall be of approved modern type, either four-wheel drive or tandem type having sufficient capacity to efficiently perform all work that may be required on the contract. Unless otherwise approved by the Ministry Representative, all graders shall meet the criteria listed in Table 521-A.

Table 521-A: Criteria for Blade Graders

Minimum Weight	7.25 t
Minimum Blade Length	3.6 m
Minimum Wheelbase	5.5 m
Minimum Cubic Displacement of Pistons: 4 Cycle Motors	7.0 L
Minimum Cubic Displacement of Pistons: 2 Cycle Motors	4.0 L

521.23 Pulverizing and Rotary Mixing Machines – Pulverizing and rotary mixing machines to be used on the work shall be standard machines, factory produced by a reputable manufacturer, so designed and constructed as to satisfactorily break up all base aggregations into constituent particles and capable of thoroughly and uniformly mixing the granular base material and applied asphaltic binder so as to produce a uniformly coated mix free from lumps, balls or segregation. The Contractor may use any machine of the Contractor's choice, subject to the approval of the Ministry Representative.

521.24 Compaction Equipment – Compaction equipment shall consist of approved steel wheel rollers, pneumatic tired rollers, vibrating rollers or tampers.

521.24.01 Self-Propelled Steel Rollers – Self-propelled steel rollers may be three-wheel rollers or tandem rollers. These rollers shall be equipped with power units of not less than four cylinders, unless otherwise approved by the Ministry Representative, and under working conditions shall be capable of developing a compression in the rear wheels of not less than 4.0 kg/mm² over a minimum roller width of 500 mm. At least one roller on each job shall be capable of developing a roll pressure of 4.5 kg/mm² of width. Rollers shall be in good working condition and free from backlash, faulty steering mechanism or worn parts. Rollers shall be equipped with adjustable scrapers to keep the rolls clean and with an efficient means of keeping the wheels wet to prevent mixes from sticking to the rolls. Rolls shall be free of flat areas, dents, openings or projections, which will mar the surface of the pavement.

521.24.02 Pneumatic Tire Rollers – Pneumatic tire rollers shall be either self-propelled or towed type, single or double axle, having a minimum effective rolling width of 1.2 m. The rollers shall be equipped with smooth tread pneumatic tires of equal size and diameter. The wheels of the roller shall be so spaced that two passes of a single axle roller or one pass of a two-axle roller will accomplish one complete coverage equal to the rolling width of the machine. There shall be a minimum 6 mm overlap of the tracking wheels of a double axle roller. The wheels of towed rollers to be used on granular base preparation may wobble. The roller shall be so constructed that the contact pressure shall be uniform for all wheels and the tire pressure of the several tires shall not vary more than 35 kPa. Pneumatic tire rollers shall be constructed with ample ballast space to provide an operating weight of 4.5 to 8 kg/mm² of tire track width. The total operating weight of the roller may be varied by the Ministry Representative. The towing vehicle for towed type roller shall be equipped with smooth tread pneumatic tires.

521.24.03 Vibrating Tampers – Vibrating tampers to be used for the compaction of bituminous base in places inaccessible to rollers, shall be of a type suitable for the work to be performed and shall be subject to the approval of the Ministry Representative.

521.24.04 Vibrating Compaction Equipment – The Contractor may use vibrating compaction equipment to compact the stabilized base, provided that such equipment produces compaction and surface finish equal to that obtainable by using a steel wheel finish roller having a minimum compression roll pressure of 4.5 kg/mm².

CONSTRUCTION

521.31 Methods – Based on whether the granular base has been completed by others or by the Contractor, the applicable construction method specified below shall be followed.

521.31.01 Method I –Where the granular base has been previously completed by others, Method I shall apply as follows.

(a) The granular base shall be regraded to true line, grade and cross section; the top 65 mm of the base shall be loosened by means of scarifying. The lumps of bonded material are to be broken to constituent gravel by harrowing, blading or rotary tilling. The loose material is to be bladed into a windrow, approximately 10 cm in height on both sides of the road. This windrow will act as a dam in case of spillage or the occurrence of rain after the spraying has taken place.

(b) The prepared gravel surface shall be primed on the side not obstructed by the windrow, to the outside width required at a rate of 1 L/m2 to 1.5 L/m² or as may be ordered, employing the bituminous binder selected by the Ministry Representative within a spraying temperature range which corresponds to a kinematic viscosity of 150 mm²/s to 200 mm²/s in the binder.

Priming shall be applied only when the surface is dry or slightly damp, and unless otherwise permitted by the Ministry Representative, when the air temperature in the shade is not less than 10°C rising or 12°C falling. When the prime has been completely absorbed by the prepared surface, blade the windrow of loose surface course onto the primed half of the road and prime the untreated half of the surface in a like manner.

Generally, the applicable provisions of SS 502.21 shall govern this work.

(c) Immediately prior to the application of bituminous binder, the windrow of loose gravel shall be bladed across the road to a full width uniform cross section. If wet or damp, it shall be first bladed back and forth until dry unless an emulsion is to be used. Cutback bituminous binder shall not be applied until the moisture content has been reduced to 1.5%. In the case of emulsions, mixing may commence when the moisture content of the gravel has been reduced to 4%.

No asphaltic binder shall be sprayed on the granular aggregate unless the average temperature of the aggregate on the road is 13°C or higher and such temperature can be maintained in the aggregate until the full mixing cycle has been completed.

Upon the layer of graded aggregate, the bituminous binder shall be applied uniformly at a rate of 1.35 L/m², at an application temperature required to produce a kinematic viscosity of 150 mm²/s to 200 mm²/s in the binder.

A disc harrow or rotary type mixer shall immediately follow the distributor after the application of the bituminous binder and shall continue to operate on the treated strip until all free bituminous material has been mixed into the mineral aggregate. The treated mixture shall then be further mixed with a blade grader until a uniform mixture results.

The mixture shall then be spread uniformly across the road once more, a second uniform application of bituminous binder at a rate of 0.9 L/m² to 1.8 L/m² as may be directed by the Ministry Representative shall be sprayed at an application temperature to give a final asphalt content of 4% to 6% of the mix. The second application of bituminous binder shall be mixed with the mineral aggregate in a similar manner to the first application, taking care that during the blade mixing none of the untreated base below the prime or shoulder material is brought into the mix. Blade mixing shall be continued until the bituminous binder and mineral aggregates are thoroughly mixed to uniform colour, free from fat spots, balls and uncoated particles. If the mixture should become wet before the mixing process is complete, the mixing operations shall be continued until it has dried out. After final mixing, the mixture shall be brought to a single windrow.

(d) Before the finished mixture is finally spread for compaction, a triangular cut shall be made with a blade at each edge of the base course to provide for a thickened edge of bituminous mixture. The cut shall be approximately 50 mm deep at the outer edge and slope to zero, 0.6 m in toward the centre. In making the cut, the excavated material shall be thrown to the shoulder in a small windrow against which the mixture shall be spread. The bituminous mixture shall be spread for compaction from the large windrow. It shall be bladed from this windrow in a succession of thin layers to correct thickness and a uniform cross section with a camber of 1 in 45 (except at curves, where the correct super-elevation originally set for base shall be followed).

After the mixture has been spread as specified, it shall be compacted by means of rollers, the operation shall begin at the edges of the course and progress toward the centre parallel to the centerline of the roadway, overlapping in successive passes by at least one-half the width of the roller. Initial breakdown compaction may be obtained by use of steel or rubber tired rollers until no appreciable surface movement or roller marks are visible and the surface has been satisfactorily sealed. Final rolling to eliminate rubber tire or other marks and irregularities shall be carried out using an

approved steel wheel finish roller or steel wheel vibrating rollers or other satisfactory equipment.

Compaction shall be carried out to the satisfaction of the Ministry Representative.

Should unsatisfactory areas develop during compaction, they shall be corrected as directed by the Ministry Representative.

521.32 Method II – When the granular base and base stabilization are included in the Contract, Method II shall apply as described hereunder:

(a) The first lift of granular base, 90 mm, shall be hauled to the roadway, spread, graded to line, grade and cross section, and compacted to 100% of the Standard Proctor Maximum Dry Density as obtained in the laboratory following ASTM D698.

(b) On completion of the above roadways, as described in the foregoing, priming may be commenced.

(c) On completion of priming and curing the second lift, 60 mm of granular base may be hauled to the primed roadway and spread in a uniform windrow and mixing commenced as described in SS 521.31.01(c)

(d) On completion of base stabilization, the remaining granular base may be applied, i.e., from the edge of the stabilized base to the shoulder.

MEASUREMENT

521.81 Spray Asphalt – Spray Asphalt will be measured by the LITRE sprayed on the road.

521.82 Process Base Gravel – Process Base Gravel will be measured by the SQUARE METRE of completed surface, constructed to dimensions ordered.

PAYMENT

521.91 Payment – Payment for asphalt stabilizing granular base course in place will be made as follows:

(a) **Spray Asphalt** – Payment for Spray Asphalt will be at the Contract Unit Price per litre for the actual number of litres sprayed on the road converted to its volume at 15.6°C. The rate for spraying asphalt shall cover acceptance of asphalt at points of delivery, storing, heating, transport, spraying, traffic control, labour superintendence, equipment and machinery required to carry out all asphalt spraying work including priming described in this Section.

(b) **Process Base Gravel** – Payment for Process Base Gravel will be at the Contract Unit Price per square metre of measured completed surface, constructed to dimensions ordered. Payment will not be made for extra width of surface mixed or compacted beyond the limits specified.

The rate for processing base gravel shall cover provision of all tools, labour, equipment and services required to cut, shape, dry, pulverize, mix and lay out the base; all compaction; warning signs, barriers, lights, traffic control and superintendence required to thoroughly mix and compact the asphalt and aggregate in place on the road.

Payment of measured quantities at the bid rates for the above two items shall be full compensation for all work and services performed and materials and equipment supplied to complete asphalt stabilization of granular base course material according to the description and intent of this Section.

SECTION 531
ASPHALT SURFACE TREATMENTS

DESCRIPTION

531.01 Scope – This Section specifies the materials, plant, equipment and quality of work required for the construction of asphalt surface treatments with cover aggregates.

531.02 Description of Work – Surface treatment shall consist of the application of a selected asphalt binder material to a previously constructed bituminous pavement, or a well compacted crushed granular surface followed by the application and embedment of a course(s) of selected aggregate to provide a new surface which shall be impervious to moisture, have a non-skid texture, high luminosity and good riding qualities. This normally would be a graded aggregate seal or a coarse sand seal or any other surface treatment outlined in the Special Provisions for the Project work of the Contract.

531.03 Types of Graded Aggregate Seals – There are six types of graded aggregate seals as characterized by the maximum and minimum gradations of the cover aggregate as defined in SS 531.17.

The Classes and their common applications are:

(a) **Class A** – Double seals on gravel highways

(b) **Class B** – Double seals on gravel highways, and single seals low traffic paved roads

(c) **Class C and Mod C** – Major Highways

(d) **Class D** – Major Highways

(e) **Class E** – Residential areas and second seals

MATERIALS

531.11 Asphalt Materials – Asphalt(s) to be used on surface treatment may be standard rapid curing cutback asphalt; medium curing cutback asphalts; or anionic or cationic emulsions including high float emulsified asphalts meeting the requirements of SS 952. Types and grades of asphalt binder proposed for use on the Project will be specified in the Special Provisions.

531.12 Coating and Anti-Stripping Additives – For general use or for experimental or trial purposes, the Ministry Representative may direct the use of surface active "coating and anti-stripping additives" or any other materials formulated to improve the coating and adhesive properties of the asphalt binder to the selected cover aggregate. In such cases, instructions concerning the type of additive to be supplied and the method of incorporation with the asphalt binder will be specified in the Special Provisions or, if the work has commenced, will be the subject of supplementary written instructions to the Contractor.

531.13 Mineral Aggregate Supply – Mineral cover aggregate meeting the requirements described hereunder shall be supplied by the Contractor unless specifically described otherwise in the Special Provisions.

531.14 Mineral Aggregate – Method of Test – Mineral Aggregate will be tested in accordance with the ASTM tests listed in Table 531-A.

Table 531-A: Test Methods for Mineral Aggregates

ASTM Test	Name of Test
C88	Standard Test Method for Soundness of Aggregates by use of Sodium Sulfate or Magnesium Sulphate
C117	Standard Test Method for Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by WashingWash Test of Aggregates.
C127	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregates
C128	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Fine Aggregates
C136	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates Sieve Analysis of Aggregates.
C142	Standard Test Method for Clay Lumps and Friable Particles in Aggregates.
C566	Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying
D75	Standard Practice for Sampling Aggregates.
D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
D4791	Standard Test Method for Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate
D5821	Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
D6928	Standard Test Method for Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus.

531.15 Mineral Aggregate – Physical Requirements

531.15.01 Composition – Mineral cover aggregate shall consist of clean, sound, hard, durable particles or fragments of sand, gravel or crushed stones or combination thereof; containing a minimum of thin elongated or flaky pieces such as may cause the height constant of the particles to be less than 0.6. It shall be substantially free from loosely bonded aggregations, clay lumps or other objectionable matter and shall not, in the Ministry Representative's opinion, be markedly hydrophilic in character.

531.15.02 Test Requirements – For the purpose of testing, the mineral cover aggregate shall be considered to be divided into coarse and fine (sand) fractions. Fine aggregates shall consist of all material passing through a 4.75 mm standard sieve, unless some other boundary shall be specifically described under the respective Method of Test.

The mineral cover aggregate, when tested by the methods listed in SS 531.14, shall meet the following criteria:

(a) **Clay Lumps –** Coarse aggregate shall not contain more than 0.5% by mass of clay lumps or other aggregation of fine material.

(b) **Absorption –** maximum 1.0% by mass for coarse aggregate; maximum 1.5% by mass for fine aggregate.

(c) **Soundness –** When tested for soundness by ASTM C 88, the aggregate shall not show a weighted loss for five cycles in excess of 12% when Magnesium Sulphate is used (or have a satisfactory record of service).

(d) **Degradation –** No individual test shall have more than 20 when tested in accordance with ASTM D 6928.

531.16 Not Used

531.17 Graded Aggregate Seal – Requirements

531.17.01 Gradation – Gradations shall meet the limit stated in Table 531-B for each Class of graded aggregate seal.

(a) **Fracture** – When tested according to SS 202, Appendix 1 fractured faces, the minimum % by mass on coarse aggregate shall be as follows:

(i) Classes A & B: 70%

(ii) Classes C, D, & E: 85%

(b) **Variation Limits** – Once the grading curve is established within the above limits, the maximum permissible variation of the mean of any five consecutive tests from the grading curve shall be within the tolerance specified in Table 531-C.

These tolerances do not waive the requirement that the running average of five consecutive tests must be maintained at all times inside the limits specified in SS 531.17.01. Should the Contractor not be able to maintain the grading within the variation limits specified above they shall construct a second stockpile. This does not alleviate the Contractor from consistently maintaining the grading curve within the tolerances indicated in the Gradation Limits below.

Table 531-B: Gradation Limits for Graded Aggregate Seal

Sieve size (mm)	% Passing by Mass by Class					
	A	B	C	Mod C	D	E
19	100	-	-	-	-	-
16	-	100	100	100	-	-
12.5[1]	60-90	60-90	-	-	100	-
9.5	40-80	35-75	30-70	25 -55	30-80	100
4.75	20-60	15-50	25-45	7 – 30	25-45	0-30
0.600	0-25	0-15	5-20	0 -10	5-20	0-10
0.075	0-7	0-5	0-3	0 – 3	0-3	0-3
Ratio[2] (9.5/4.75)	-	-	-	1 – 1.5	-	-

Notes:

[1] The Contractor may use either the 12.5 mm or the 13.2 mm sieve, at the Contractor's option.

[2] The "Ratio" is the ratio of the mass of material retained on the 9.5 mm sieve, divided by the mass of material retained on the 4.75 mm sieve.

Table 531-C: Variation Limits

Sieve Size (mm)	Maximum Permissible Tolerance % Passing by Mass (ASTM C117 & ASTM C136)
4.75 to 19.0	± 4.5
0.600	± 3.5
0.075	± 1.0

531.17.02 Coarse Aggregate

(a) Defined as all material retained on the 4.75 mm sieve.

(b) Consist of crushed stone or gravel, or a combination thereof; or materials naturally occurring in a fractured condition; or materials naturally occurring of highly angular nature or rough texture with a maximum result of 20 when tested in accordance with ASTM D6928.

(c) Free from coating of clay, silt or other deleterious material and shall meet the applicable test requirements of SS 531.14 and SS 531.15.

531.17.03 Fine Aggregate

(a) Defined as all materials passing the 4.75 mm sieve and shall meet the applicable test requirements of SS 531.14 and SS 531.15.

(b) Clean, hard, durable, moderately sharp, and free from coatings of clay, silt or other deleterious material and shall contain no clay balls or other aggregations of fine material.

531.17.04 Mineral Filler and Mineral Dust

(a) Mineral filler shall consist of all mineral matter passing the 0.600 mm sieve and the mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

(b) Mineral filler and mineral dust shall be free from organic matter.

(c) Mineral filler shall be non-plastic when tested in accordance with ASTM D4318.

531.18 Presealing Sand – Sand required for use as a fine aggregate in connection with presealing operations conducted under SS 531.36 shall meet the physical requirements of SS 531.15, but unless otherwise specified by the Ministry Representative, shall have a grading as specified in Table 531 D.

Table 531 D: Physical Requirements for Sand in a Fine Aggregate

SIEVE SIZE (mm)	PERCENTAGE PASSING
4.75	100
2.36	80 – 100
1.18	60 – 90
0.300	25 – 70
0.150	10 – 40
0.075	2 – 12

Note: All aggregate passing the 0.425 mm sieve shall be non-plastic.

EQUIPMENT

531.21 General

(a) **Equipment List –** A minimum of fourteen (14) days before commencing work under the Contract, the Contractor shall submit a list of equipment available and ready for use on the Project.

(b) **Equipment Inspection –** Before any work is permitted to commence, the Contractor's equipment and plant will be carefully inspected and should any of it fail to meet the required standards, work will not be permitted to commence until such deficiency is corrected.

(c) **Equipment Release –** Provision of all equipment listed shall become an essential part of the contract and no equipment or plant may be released from the Project work for use elsewhere until completion, except by the express written permission of the Ministry Representative.

(d) **Access and Adjustment –** The Ministry inspectors shall have access at all times to any plant, equipment or machinery to be used on the Project in order to check calibration, control or operating adjustments. Where adjustment appears necessary, it shall be done by a competent operator or mechanic or instrument repair technician, as required.

(e) **Minimum Equipment –** Plant, machinery and equipment specified hereafter shall represent a minimum requirement for the type of machine described and does not represent a comprehensive list of equipment to be supplied by the Contractor who shall supply whatever plant and equipment may be necessary to the proper and efficient completion of the work

(f) **Skilled Operators –** All plant and equipment shall be under the control of experienced and skilled operators.

531.22 Pressure Distributor – The pressure distributor used for applying asphalt materials shall consist of a fully insulated tank with a minimum capacity of 5,000 L, permanently and rigidly mounted on a truck or fifth wheel trailer provided with pneumatic tires and propelled by a power unit capable of maintaining a constant and uniform speed.

The pressure distributor, to be acceptable, shall be provided with at least the following equipment:

(a) A quick opening gate in the dome.

(b) A dipstick calibrated in 20 mm increments and a calibration chart showing the volume for each 20 mm of depth, or provide the manufacturer dipstick and chart. Readily visible external depth gauge indicating the quantity of asphalt in the tank to the nearest 50 L.

(c) A means of heating the asphalt material uniformly to any temperature up to 175°C and maintaining the contents constantly at any selected temperature without any local overheating.

(d) A heavy duty, industrial-type thermometer and a remote reading dial thermometer.

(e) Rear-mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment. The spray bars shall be capable of adjustment to provide a uniformly sprayed surface from a minimum of 0.6 m to at least 4 m width.

The spray bars shall be capable of producing up to triple coverage at one pass. The feed to the spray bars must be arranged so that there is a constant circulation

of hot asphalt material to the bars to ensure a uniform viscosity and constant pressure of the liquid asphalt at each nozzle both before and during spraying operations.

(f) The feed manifolds to the spray bars shall be provided with positive filtration to prevent clogging of bars and nozzles.

(g) Rotary adjustable spray bar nozzles shall be of the same type, correct size and manufacture; and set to produce uniformly fan-shaped sprays without atomization. Nozzles shall be inspected daily and any nozzle having nicked or damaged edges shall be replaced.

(h) Spray nozzles shall be provided with valves capable of instant full opening and positive cut-off.

(i) A connection to a circulating pressure manifold to which a hose may be attached for a single nozzle hand-operated outlet.

(j) A pressure pump capable of applying the asphalt material at a minimum rate of 180 L per minute, per metre of spray bar.

(k) A pressure gauge accurate to within 15 kPa and a metre calibrated in litres per minute, shall be readily visible.

(l) A fifth-wheel tachometer, calibrated in metres per minute and readily visible to the operator of the distributor.

(m) Vehicle spring "tie-downs" to prevent the change in height of the spray bars due to the reduction of the asphalt load during spraying.

(n) Electronic distance measuring device, with print out connected to the transmission, that is capable of measuring each spray length and the total length sprayed during the day.

531.23 Mechanical Spreader – Cover aggregate shall be spread by means of an efficient, self-propelled spreader mounted on pneumatic tires and capable of continuously and uniformly spreading closely regulated quantities of crushed aggregates at the application rates selected. The loading hopper shall be of such a capacity as to ensure continuous operation between dumps of aggregate. Aggregate shall be transferred from the loading hopper to the front spreading device in such a manner as to ensure a uniform flow and prevent segregation of particles.

The spreading mechanism shall be so designed as to distribute and spread aggregate across the selected width without segregation. Aggregate shall be applied to the freshly sprayed surface by use of a screen so that the largest particles are first placed on the road with the smaller fractions falling on top. The approved screen is to be mounted beneath the discharge openings of the spreader mechanism in such a manner that the angle of the screen relative to the road surface can be easily and quickly changed.

Suitable provision shall be made to prevent aggregate from rolling on the freshly sprayed surface. The cover aggregate shall be applied ahead of the spreader wheels, with sharply defined, straight edges and without any lateral movement of the aggregate after initial application.

A walkway with protective railings, conforming to the applicable Occupational Health and Safety Regulations, shall be provided at the front of the spreader so that the hopper operator has safe access to the individual gate control levers.

The spreader shall be designed to apply aggregate at any application width between 2 m and 4.12 m. *Note: A 4.26 m hopper is required.*

531.24 Rollers – Self-propelled pneumatic tired rollers (mass range from 13,600 kg to 19,000 kg) shall be used on the work. The rollers shall have a minimum effective rolling width of 1.83 m. The rollers shall be equipped with smooth, wide-tread compaction tires of equal size and diameter. The wheels of the roller shall be spaced so that one pass of the roller shall accomplish one complete coverage equal to the rolling width of the machine. There shall be a minimum overlap of 6 mm of the tracking wheels of the roller. The wheels shall be so mounted as to oscillate in pairs. The roller shall be provided with ample ballast space and be so constructed that the specified wheel load can be uniformly applied to all the wheels, to provide similar contact pressure under all wheels when rolling on a level pavement.

The tires shall have a minimum of 16 ply and their inflation pressure shall not normally exceed the specified pressure by more than 35 kPa and shall at no time be less than specified. Inflation pressure of all of the tires shall be controlled from a central location while the machine is in motion and capable of varying the tire pressure from 206 kPa to 827 kPa. Rollers shall be ballasted and operated to the manufacturers recommended specifications.

The Ministry Representative may approve the use of compactors equal to but differing from the above requirements provided the Ministry Representative is satisfied that equally effective compaction will be obtained by use of the alternative equipment.

The Ministry Representative may approve the use of rubber-covered, vibrating drum rollers (minimum width of 1.52 m and mass of 3000 kg) or tandem steel wheel rollers in the static mode provided the drum or drums can contact the road surface over the entire width of the drums (no bridging). On shoulder seals, the use of pulled wobbly wheel rollers may be used at the discretion of the Ministry Representative. They also may be used on graded aggregate seal on gravel surfaces, where the road is wide enough to permit units to turn without damage to the road surface, sealed or otherwise.

All rollers shall be provided with equipment for spraying water continuously on all tires or wheels, while rolling is in progress, and also be provided with such means as may be required to keep the tires clean and free of adhering asphaltic binder.

531.25 Power Brooms – Brooming work shall be carried out using rotary power brooms mounted on self-propelled, pneumatic tired tractor units. The brooms shall be capable of vertical and horizontal angular adjustment and must be provided with equipment for spraying water, as required, on the surface to be broomed while the equipment is in operation. The brooms shall have sufficient power and brushing capacity to completely clean the surface of the standard lane to be treated or which has been treated within three coverages.

To be read in conjunction with SS 145.17 and SS 165.

531.26 Blade Graders – Self-propelled blade graders mounted on pneumatic tires shall be of approved modern type, either four wheel drive or tandem type, having sufficient capacity to efficiently perform all work that may be required on the contract.

531.27 Water Truck – Water trucks shall have a minimum capacity of 9,000 L. Water shall be applied from a distributor of the pressure type, equipped with spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water accurately and uniformly. Splash plate type of distributors or those equipped with spray bars that eject fine streams of water will not be permitted.

The distributor must be provided with a satisfactory means for accurately measuring the quantity of water sprayed. If the Ministry Representative so requires, measuring equipment shall be calibrated under the Ministry Representative's inspection. The Contractor shall make all necessary arrangements for obtaining water at the Contractor's own expense. In addition, the water truck shall be equipped so that it may be used as a flush truck with front-mounted, adjustable spray nozzles that produce a fan-shaped spray in the horizontal mode and can be adjusted in vertical or horizontal direction (ball-mounted). The delivery of water should be variable to single or multiple nozzles so that the desired flushing can be achieved. The water truck shall be equipped with a loading pump capable of pumping 1360 L or more of water per minute.

The spraying and flushing controls must be in the cab and accessible to the driver while the unit is in motion. The minimum spray bar width will be 2.40 m. Gravity single jet or drip bar equipment unit will not be permitted.

531.28 Sign Truck – Truck(s) of sufficient capacity to store and transport all necessary signs, stands, barriers, flashing lights, batteries, flood lights, generators, high-level warning devices, project signs, etc., is required.

531.29 Tanks, Mobile Asphalt Heating, & Storage – Insulated liquid asphalt storage tanks should meet the following minimum criteria:

(a) A positive high-low heating system, 10°C to 200°C, with a hot oil/steam circulation system.

(b) Automated heating and control system with safety shutdown.

(c) Intake/Outlet, two, each with screens installed.

(d) Positive locking valve on discharge line.

(e) Sampling valve or spigot.

(f) Pyrometer; Multi point 20°C – 200°C.

(g) Direct reading levelometer.

(h) Manholes, at each end, with ladders.

(i) Tank(s) with sufficient capacity to hold one day's production requirements.

CONSTRUCTION

531.31 Temperature and Weather – Asphalt binder and aggregate shall be applied only during the hours of daylight when the shade air temperature is at least 10°C for graded aggregate seals and is rising; and when the surface of the road is dry and at a minimum temperature of 20°C and rising. Operations shall cease when the road temperature falls below 20°C for chip seals and 15°C for graded aggregate seals. No application shall take place when, in the opinion of the Ministry Representative, the weather or roadbed conditions are unfavourable, i.e., weather is misty or rainy, precipitation is predicted for the construction area within 12 hours or the atmospheric temperature at the construction area of less than 12°C is predicted within 24 hours. No chip seal shall be applied before June 1 or after August 15.

531.32 Traffic Control – Traffic control shall meet the requirements of SS 194, the Special Provisions and the following subsections.

531.32.01 Signs and Barriers – The Contractor shall provide all warning and instructional signs, barriers, lights, electronic communicating devices and flagpersons required to properly and safely control all traffic around or through the work in accordance with SS 194, and as may be specially required below.

531.32.02 Detours – Where conditions indicate that traffic may be detoured, the Contractor shall arrange to close off the section of road to be sealed and erect detour and route signs. After the completion of the sealing operations traffic shall not be allowed onto the closed sections, for the periods specified in SS 531.32.04, SS 531.32.06, and SS 531.37.04, unless otherwise directed by the Ministry Representative.

531.32.03 Shoulder Detours – Where shoulders are wide and firm enough to accommodate traffic that cannot be detoured along an alternative route, the Ministry Representative may direct that traffic be detoured onto the shoulder. If required, and approved by the Ministry Representative, the shoulder shall be made up, rolled and graded on a force account basis. A designated liquid asphalt shall be sprayed from the edge of the road to the edge of the shoulder, as required, and allowed to cure before traffic is detoured off the road to be sealed and onto the shoulders.

The Contractor shall provide sufficient pilot cars and barriers to maintain traffic movement at a maximum speed of 40 km/h solely on the shoulders, during the course of work. Once a section has been sealed, traffic will be regulated as indicated in SS 531.32.04.

531.32.04 Traffic Through Work – Where it is not possible to detour traffic, sealing work shall normally be carried out in lane widths and traffic shall be regulated by pilot cars on the opposite traffic lane to that on which construction is in progress. Traffic travelling on the old road surface may be permitted to travel up to a maximum speed of 50 km/h consistent with safety requirements. Traffic shall only be allowed to travel on the newly-sealed lane under control of pilot car(s) which shall enforce a 20 km/h speed limit up to 4 hours and a 30 km/h speed limit for a period of up to 24 hours, as determined by the Ministry Representative, after completion of the sealing operations; except in the case of pre-seal construction when the period of control shall be up to 4 hours after completion of the pre-sealing operation.

In the event of prolonged inclement weather, traffic control will be maintained as long as the Ministry Representative deems necessary to protect newly laid surface seals.

531.32.05 Pilot Cars – Sufficient pilot cars shall be provided so that the delay to traffic shall be no greater than 20 minutes. They shall be equipped in accordance with SS 194.

531.32.06 Communications – The flag people, pilot car(s), sign truck, spreader and supervisor's vehicle shall have an effective electronic means of communication so that instructions and information can be quickly and accurately relayed over the total length of the control area.

531.32.07 Flood Lights – During periods when headlights must be used, including during daytime periods of poor visibility and high dust conditions, two self-contained floodlights shall be used to illuminate each of the flagging stations. These floodlights shall be of at least 1000 watts and be complete with reflector, lens and adjustable stand at least 2.15 m high. The light, reflector and lens must be mounted so that they are capable of both vertical and horizontal adjustment. The stands are to be sufficiently sturdy so that they can withstand a 30 km/h wind without the use of ballast. The generators shall have sufficient capacity to enable them to run one shift under full load.

531.32.08 Flag Personnel – Flag personnel shall be equipped in accordance with SS 194 and provided as required to safely control and conduct traffic through the job.

531.33 Preparation of Roadway – The patching and levelling of the pavement to be sealed may be part of a sealcoating contract or may be effected by others.

531.34 Sweeping – Sweeping shall meet the requirements of SS 145, SS 165, and SS 531.27 regarding equipment for watering and the following:

(a) Immediately prior to the commencement of the sealing operations, all loose aggregates, dust, dirt, caked clay or foreign materials shall be removed from the width of the surface to be treated by brushing with power brooms supplemented by hand push brooms, shovels or the use of a power grader, and where necessary by flushing.

(b) Particular care shall be taken to thoroughly clean to the outside edges of the strips to be treated and to ensure that the sweepings are not so deposited on the shoulder as to permit subsequent contamination of the treated surface.

531.35 Patching – In order to avoid delays to the Contractor, any patching not done, and in the opinion of the Ministry Representative required before sealing operations begin, may be done by the Contractor as directed by the Ministry Representative with materials supplied by the Ministry and paid for under an Order for Extra Work.

531.36 Pre-Seal Treatment

531.36.01 Description – Wherever the pavement is in a worn, open, dry or porous condition, or on gravel surfaces which, in the opinion of the Ministry Representative, would result in high absorption of the asphalt binder, the surface shall be pre-sealed. All pre-sealing work on a Contract shall, unless otherwise ordered by the Ministry Representative, be fully completed prior to the commencement of seal coat work.

531.36.02 Application of Asphalt Pre-Seal – After the surface has been prepared in accordance with SS 531.33 and SS 531.34, a selected asphalt binder shall be sprayed, at such application rates as directed, so that just sufficient asphalt material shall be applied to the most absorptive sections of the mat, avoiding an excess elsewhere. The asphalt binder shall be applied at the temperature at which the Saybolt-Furol Viscosity of the material is in the range of 50 to 75 seconds.

531.36.03 Application of Sand Cover –Application of sand in sufficient quantity to completely blot up all excess unabsorbed asphalt material shall follow within five minutes or as may be directed by the Ministry Representative.

At the time of spreading, the sand shall not contain more than 5% moisture by mass.

531.36.04 Rolling – Rolling shall commence immediately after the application of cover sand and shall be completed within 15 minutes. Sufficient pneumatic tired rollers shall be used to ensure that the entire surface of the treated road receives complete coverage within the specified time, at a rolling speed not in excess of 8 km/h.

531.36.05 Bleeding – Bleeding spots or areas shall receive a further application of sand blotter and be re-rolled. This process shall be repeated until no further bleeding occurs.

531.36.06 Traffic – No traffic shall be permitted on pre-sealed sections for up to four (4) hours or as the Ministry Representative may direct.

531.36.07 Curing Time – No seal coating work shall be carried out on the pre-sealed surface until 24 hours after completion of the pre-seal or as the Ministry Representative may direct.

531.36.08 Sweeping – Prior to commencement of the main seal-coat treatment all loose, unbonded sand shall be swept from the pre-sealed surface in accordance with SS 531.34.

531.37 Single Pass Surface Treatment

531.37.01 Application of Asphalt Material – The selected asphalt material shall be uniformly sprayed on the properly prepared surface at a rate of 0.5 to 2.75 L/m^2, or as may be directed, and at such temperature as shall cause the Sayholt Furol Viscosity of the material to be within 50 to 75 seconds.

Wherever it is possible to detour traffic, the surface may be sprayed full width at one pass of the distributor. Two spreaders must be used.

Normally, where it is not possible to detour traffic only one half of the roadway shall be sprayed at one time, permitting one-way controlled traffic on the other half. In no case shall a longitudinal joint be so placed that it is in a wheel path, the exception being when a double graded seal is being applied.

In order to ensure a uniform distribution of asphalt material and avoid the formation of a lap or ridge at the transverse junction of two successive applications, building paper shall be spread over the treated surface for a sufficient length back so that the spray jets are operating properly (fully open) when the uncovered surface is reached. The building paper shall then be removed and disposed of prior to the application of aggregate (Refer to SS 165). There shall be a maximum 100 mm overlap of sprayed material at the longitudinal joint between sprayed lanes.

The spray shall be cut off before the distributor tank has been completely emptied to ensure that a full application of asphalt material is made up to the point of cut off. Pressure hand sprays shall be used to properly treat small patches or inaccessible places that have been missed by the distributor. Care shall be taken and building paper shall be used, if necessary, to ensure against staining any curbs, guardrails or faces of any improvements near the work.

The progress of the application of the asphalt material shall be governed by the weather, the curing rate of the asphalt and the rate at which the cover aggregate is applied and rolled. No traffic shall be permitted on the uncovered asphalt material. The Contractor shall organize work so that seal-coating can normally be completed to the full width of the road surface each day.

531.37.02 Application of Cover Aggregate

(a) Application – The uniform application of the specified cover aggregate, at the spread rate set by the Ministry Representative, shall immediately follow the application of the asphalt material, before the emulsion breaks or within 2 minutes if a cutback asphalt is sprayed. An approved screen shall be used to place the larger chips or particles of aggregate onto the sprayed asphalt first so that maximum embedment of the larger material is obtained. Any segregation shall be corrected immediately.

(b) Moisture Content Graded Seal Aggregates – At the time of spreading, the moisture content of the aggregates shall be such that the material will flow freely in the hopper without segregation and/or a build-up of material on the screen and shall be less than the maximum allowable in Table 531-E.

(c) Correction of Non-Uniformity – Prior to and during the rolling operation, any non-uniformity in the distribution of the cover aggregate shall be corrected by spreading additional aggregate and/or by light raking or hand brooming over such sections of the road as the Ministry Representative directs.

Table 531-E: Maximum Mass Percentage for Moisture Content of Cover Aggregate

Aggregate Types	Maximum Allowable Moisture Content for Liquid Cut Back and Asphalt Emulsions
Fine Cover Sand (≤ 4.5 mm)	5%
Coarse Cover Sand (>4.5 mm)	4%
Small Cover Aggregate (< 9 mm)	3%
Medium Cover Aggregate (9 mm – 12 mm)	2%
Large Cover Aggregate (> 12 mm)	1%

531.37.03 Rolling – Immediately after the aggregate has been uniformly spread, the road surface shall be rolled sufficiently to embed the aggregate into the asphalt material to the satisfaction of the Ministry Representative.

Sufficient pneumatic tired rollers shall be used to ensure that the entire surface of the aggregate receives at least five complete coverages within 15 minutes of the application of asphalt material; or when asphalt emulsion is used, five complete coverages shall be completed before the emulsion starts to break. The speed of the rollers shall be such that the suction of the tires shall not displace the aggregate from the road surface or a maximum of 8 km/h, whichever is less.

The completed surface shall present a uniform appearance. Any segregation or non-uniformity in the distribution of the aggregate shall be corrected by spreading additional material over the area and re-rolling with five coverages of the rollers.

531.37.04 Protection of the Surface

(a) No traffic shall be permitted on the sealed surface until after the rolling has been completed and when the Ministry Representative is satisfied that the applied aggregate will not be picked up or be displaced under traffic. Generally, traffic shall not be permitted on the sealed roadway until at least 2 hours after the rolling is completed. If traffic must be passed over a sealed surface before 2 hours have passed then they will be kept to a speed of 20 km/h or less. After 2 hours, then the speed may be increased to 30 km/h up to 4 hours and not more than 50 km/h for 24 hours or until the surface has been swept. The curing period shall be increased if so directed by the Ministry Representative.

(b) From the completion of rubber tired rolling on any section of sealed surface and for a further continuous period of 3 days or more including weekends and public holidays, the Contractor shall have a vehicle(s) and maintenance crew (a minimum of three employees) immediately available to arrest any tendency of the asphalt binder to bleed through the surface of the cover aggregate and to prevent pickup and blackening of the surface by the action of traffic. All bleeding spots shall be blotted with a sufficient quantity of blinding sand necessary to correct the defect and in such a manner as the Ministry Representative shall direct. No payment will be made for the supplying and placing of the sand.

(c) Blinding Sand – In the event of rain, traffic will be kept off a newly sealed surface until sand, provided by the Contractor, has been applied in sufficient quantity to protect the surface of the road from damage. Sanding will continue as long as necessary to protect the freshly laid seal. Payment for the supply and application of this sand will be by Order for Extra Work.

531.37.05 Removal of Loose Aggregate – To be read in conjunction with SS 145 and SS 165.

(a) The Contractor shall, when directed by the Ministry Representative, sweep off all surplus aggregate and spread it neatly along the gravel shoulders. Sweeping of the road surface will normally be completed within 48 hours for graded aggregate seals. This operation shall be done while the road surface is cool and care should be exercised not to disturb the aggregate that has been retained in the asphalt. In the event that cover aggregate is being disturbed, brooming will cease until curing is complete and the cover aggregate is firmly held in place. Water shall be used in sufficient quantity to reduce dust hazard and to meet pollution control requirements.

(b) Alongside barriers and curbs and where catch basins or drainage outlets are encountered, the Contractor shall pick up and dispose of the surplus aggregate, including any pertinent cleaning of catch basins, behind the barriers or curbs or within the right-of-way, in any manner the Contractor elects, subject to the approval of the Ministry Representative.

531.38 Double Pass Surface Treatment

531.38.01 Preparation of Existing Surface – Existing surface preparation shall be performed in accordance with SS 531.33 through SS 531.36 as applicable and as detailed in the Special Provisions.

531.38.02 Surface Treatment – First Pass – The first pass of the surface treatment shall be performed in accordance with SS 531.37.01 and SS 531.37.02, using the graded seal cover aggregate specified in the Special Provisions. Rolling, Surface Protection and Removal of Loose Aggregate shall conform to SS 531.37.03 through SS 531.37.05.

531.38.03 Surface Treatment – Second Pass – The second pass of the surface treatment shall commence not less than 24 hours after completion of the first pass, subject to acceptable weather conditions as detailed in SS 531.31, and as the Ministry Representative may direct. Asphalt material shall be applied in accordance with SS 531.37.01, followed by an application of graded seal aggregate as specified in the Special Provisions and detailed in SS 531.37.02, and rolling, surface protection and removal of loose aggregate per SS 531.37.03 through SS 531.37.05.

MEASUREMENT

531.81 Surface Treatment – Surface treatment will be measured by the SQUARE METRE.

531.82 Asphalt Binder – Asphalt binder will be measured by the LITRE.

531.83 Aggregates – The aggregate will be measured by the CUBIC METRE in stockpile. The volume will be determined in accordance with SS 145.21.01.

531.84 Stockpile Bases – No measurement will be made for work and materials required for the preparation of stockpile bases, as directed by the Ministry Representative.

PAYMENT

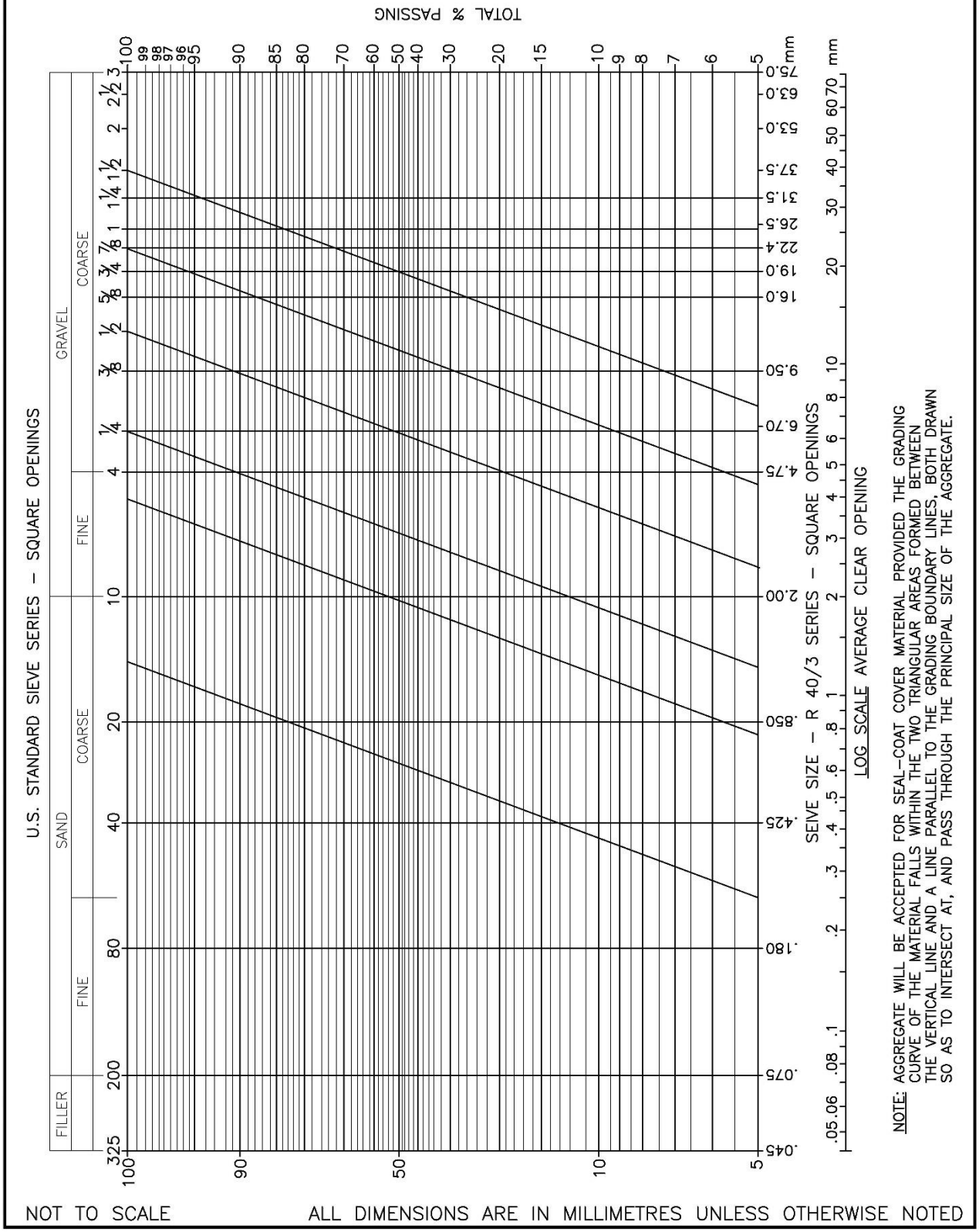
531.91 Surface Treatment – Payment for SURFACE TREATMENT will be at the Contract Unit Price per square metre. The Unit Price shall cover everything done and supplied including ordering, receiving, storing and heating of asphalt materials, traffic control equipment, and all equipment and labour required to complete the work.

531.92 Asphalt Binder – Payment for ASPHALT BINDER will be at the Contract Unit Price per litre actually sprayed.

531.93 Aggregates – Payment for AGGREGATE will be at the Contract Unit Price per cubic metre in stockpile. No additional payment will be made for washing when needed in order to meet gradation requirements.

531.94 Stockpile Bases – All work and materials required for the PREPARATION OF STOCKPILE BASES, as directed by the Ministry Representative, will be paid for, on an Order for Extra Work basis from the Provisional Sum included in the Schedule 7.

GRADING CHART FOR SEAL–COAT COVER AGGREGATES — SP531–01



NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED